

FOREST FIRE RISK MANAGEMENT

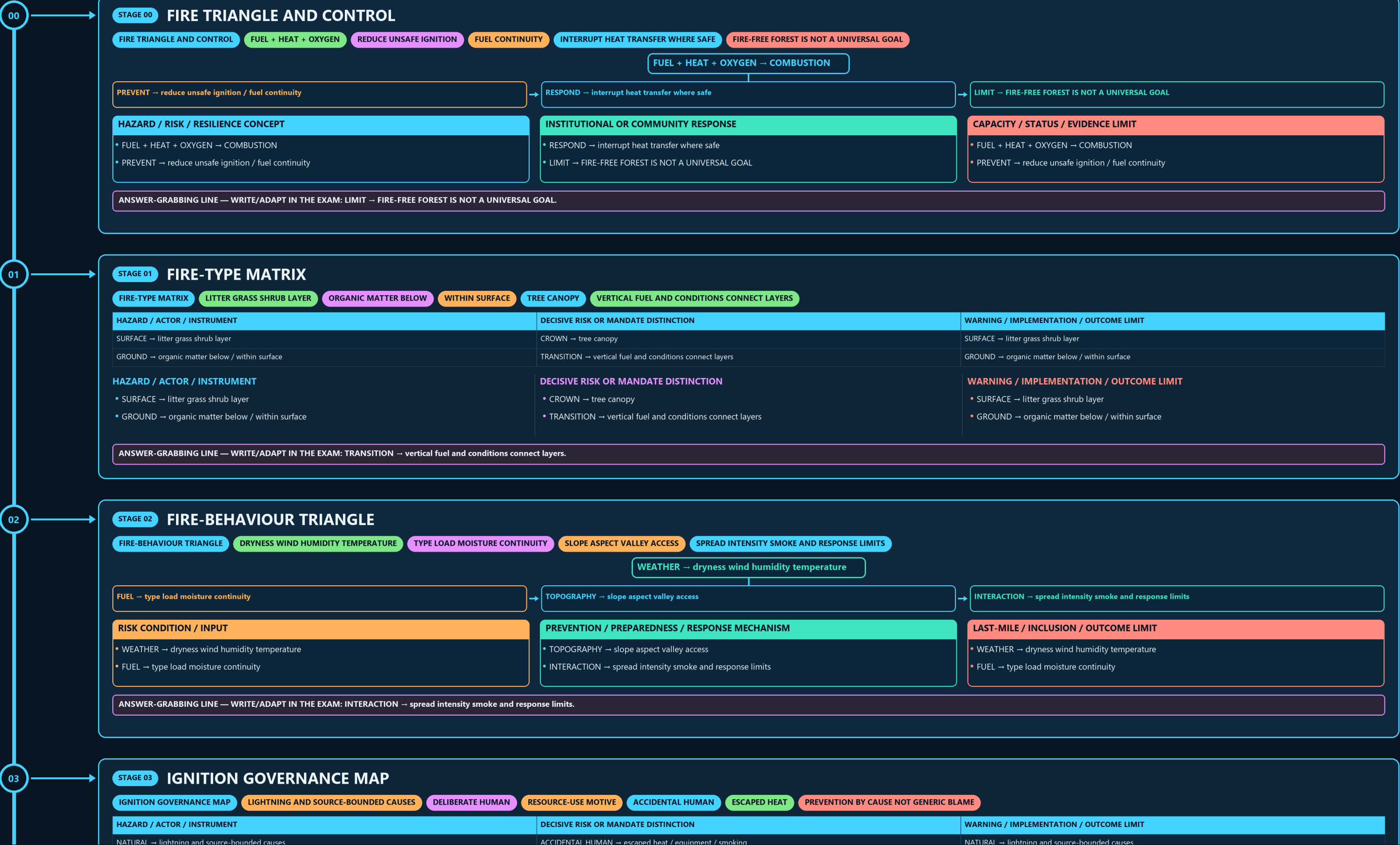
FIRE TRIANGLE AND CONTROL → FIRE-TYPE MATRIX → FIRE-BEHAVIOUR TRIANGLE → IGNITION GOVERNANCE MAP → FSI ALERT RAIL → INITIAL RESPONSE CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



RISK CONDITION / INPUT

- WEATHER → dryness wind humidity temperature
- FUEL → type load moisture continuity

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- TOPOGRAPHY → slope aspect valley access
- INTERACTION → spread intensity smoke and response limits

LAST-MILE / INCLUSION / OUTCOME LIMIT

- WEATHER → dryness wind humidity temperature
- FUEL → type load moisture continuity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTERACTION → spread intensity smoke and response limits.

03

STAGE 03

IGNITION GOVERNANCE MAP

IGNITION GOVERNANCE MAP

LIGHTNING AND SOURCE-BOUNDED CAUSES

DELIBERATE HUMAN

RESOURCE-USE MOTIVE

ACCIDENTAL HUMAN

ESCAPED HEAT

PREVENTION BY CAUSE NOT GENERIC BLAME

HAZARD / ACTOR / INSTRUMENT	DECISIVE RISK OR MANDATE DISTINCTION	WARNING / IMPLEMENTATION / OUTCOME LIMIT
NATURAL → lightning and source-bounded causes	ACCIDENTAL HUMAN → escaped heat / equipment / smoking	NATURAL → lightning and source-bounded causes
DELIBERATE HUMAN → land / resource-use motive	RESPONSE → prevention by cause not generic blame	DELIBERATE HUMAN → land / resource-use motive

HAZARD / ACTOR / INSTRUMENT

- NATURAL → lightning and source-bounded causes
- DELIBERATE HUMAN → land / resource-use motive

DECISIVE RISK OR MANDATE DISTINCTION

- ACCIDENTAL HUMAN → escaped heat / equipment / smoking
- RESPONSE → prevention by cause not generic blame

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- NATURAL → lightning and source-bounded causes
- DELIBERATE HUMAN → land / resource-use motive

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESPONSE → prevention by cause not generic blame.

04

STAGE 04

FSI ALERT RAIL

FSI ALERT RAIL

1 MODIS

SNPP-VIIRS DETECTION

2 AUTOMATED PROCESSING AND FOREST FILTER

3 SMS

MAP DISSEMINATION

4 FIELD VERIFY

HAZARD / RISK / RESILIENCE CONCEPT	INSTITUTIONAL OR COMMUNITY RESPONSE	CAPACITY / STATUS / EVIDENCE LIMIT
1 MODIS / SNPP-VIIRS DETECTION	3 SMS / EMAIL / MAP DISSEMINATION	1 MODIS / SNPP-VIIRS DETECTION
2 AUTOMATED PROCESSING AND FOREST FILTER	4 FIELD VERIFY → COMMAND → RESPONSE	2 AUTOMATED PROCESSING AND FOREST FILTER

HAZARD / RISK / RESILIENCE CONCEPT

- 1 MODIS / SNPP-VIIRS DETECTION
- 2 AUTOMATED PROCESSING AND FOREST FILTER

INSTITUTIONAL OR COMMUNITY RESPONSE

- 3 SMS / EMAIL / MAP DISSEMINATION
- 4 FIELD VERIFY → COMMAND → RESPONSE

CAPACITY / STATUS / EVIDENCE LIMIT

- 1 MODIS / SNPP-VIIRS DETECTION
- 2 AUTOMATED PROCESSING AND FOREST FILTER

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 3 SMS / EMAIL / MAP DISSEMINATION; 4 FIELD VERIFY → COMMAND → RESPONSE.

05

STAGE 05

INITIAL RESPONSE CHAIN

INITIAL RESPONSE CHAIN

VERIFY LOCATION AND ACCESS

LOCAL COMMAND + TRAINED PROTECTED CREW

TOOLS WATER COMMUNICATION AND SAFETY

ESCALATE IF CAPACITY

CONDITIONS EXCEEDED

ALERT → VERIFY LOCATION AND ACCESS → LOCAL COMMAND + TRAINED PROTECTED CREW → TOOLS WATER COMMUNICATION AND SAFETY → ESCALATE IF CAPACITY / CONDITIONS EXCEEDED

RISK CONDITION / INPUT	PREVENTION / PREPAREDNESS / RESPONSE MECHANISM	LAST-MILE / INCLUSION / OUTCOME LIMIT
ALERT → VERIFY LOCATION AND ACCESS	TOOLS WATER COMMUNICATION AND SAFETY	ALERT → VERIFY LOCATION AND ACCESS
LOCAL COMMAND + TRAINED PROTECTED CREW	ESCALATE IF CAPACITY / CONDITIONS EXCEEDED	LOCAL COMMAND + TRAINED PROTECTED CREW

RISK CONDITION / INPUT

- ALERT → VERIFY LOCATION AND ACCESS
- LOCAL COMMAND + TRAINED PROTECTED CREW

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- TOOLS WATER COMMUNICATION AND SAFETY
- ESCALATE IF CAPACITY / CONDITIONS EXCEEDED

LAST-MILE / INCLUSION / OUTCOME LIMIT

- ALERT → VERIFY LOCATION AND ACCESS
- LOCAL COMMAND + TRAINED PROTECTED CREW

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TOOLS WATER COMMUNICATION AND SAFETY; ESCALATE IF CAPACITY / CONDITIONS EXCEEDED.

06

STAGE 06

COMMUNITY PARTNERSHIP LADDER

COMMUNITY PARTNERSHIP LADDER

CONSULT ON RESOURCE-USE AND IGNITION DRIVERS

TRAIN + EQUIP + COMPENSATE

LINK EARLY REPORTING TO FOREST DEPARTMENT

DO NOT OFFLOAD STATE DUTY

CONSULT ON RESOURCE-USE AND IGNITION DRIVERS → TRAIN + EQUIP + COMPENSATE → LINK EARLY REPORTING TO FOREST DEPARTMENT → DO NOT OFFLOAD STATE DUTY

RISK CONDITION / INPUT	PREVENTION / PREPAREDNESS / RESPONSE MECHANISM	LAST-MILE / INCLUSION / OUTCOME LIMIT
CONSULT ON RESOURCE-USE AND IGNITION DRIVERS	LINK EARLY REPORTING TO FOREST DEPARTMENT	CONSULT ON RESOURCE-USE AND IGNITION DRIVERS
TRAIN + EQUIP + COMPENSATE	DO NOT OFFLOAD STATE DUTY	TRAIN + EQUIP + COMPENSATE

RISK CONDITION / INPUT

- CONSULT ON RESOURCE-USE AND IGNITION DRIVERS
- TRAIN + EQUIP + COMPENSATE

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- LINK EARLY REPORTING TO FOREST DEPARTMENT
- DO NOT OFFLOAD STATE DUTY

LAST-MILE / INCLUSION / OUTCOME LIMIT

- CONSULT ON RESOURCE-USE AND IGNITION DRIVERS
- TRAIN + EQUIP + COMPENSATE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LINK EARLY REPORTING TO FOREST DEPARTMENT; DO NOT OFFLOAD STATE DUTY.

06

STAGE 06 COMMUNITY PARTNERSHIP LADDER

COMMUNITY PARTNERSHIP LADDER

CONSULT ON RESOURCE-USE AND IGNITION DRIVERS

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CONSULT ON RESOURCE-USE AND IGNITION DRIVERS

TRAIN + EQUIP + COMPENSATE

LINK EARLY REPORTING TO FOREST DEPARTMENT

DO NOT OFFLOAD STATE DUTY

RISK CONDITION / INPUT

- CONSULT ON RESOURCE-USE AND IGNITION DRIVERS
- TRAIN + EQUIP + COMPENSATE

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- LINK EARLY REPORTING TO FOREST DEPARTMENT
- DO NOT OFFLOAD STATE DUTY

LAST-MILE / INCLUSION / OUTCOME LIMIT

- CONSULT ON RESOURCE-USE AND IGNITION DRIVERS
- TRAIN + EQUIP + COMPENSATE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LINK EARLY REPORTING TO FOREST DEPARTMENT; DO NOT OFFLOAD STATE DUTY.

07

STAGE 07 FUEL-BREAK BOUNDARY

FUEL-BREAK BOUNDARY

FIRE LINE

MAINTAINED BREAK IN FUEL CONTINUITY

PRESCRIBED BURN

AUTHORISED PLANNED FUEL TREATMENT

OBJECTIVES CONDITIONS SUPERVISION CONTINGENCY

GUARANTEED CONTAINMENT OR LICENCE FOR IGNITION

HAZARD / ACTOR / INSTRUMENT

FIRE LINE → maintained break in fuel continuity

PRESCRIBED BURN → authorised planned fuel treatment

DECISIVE RISK OR MANDATE DISTINCTION

BOTH → objectives conditions supervision contingency

NEITHER → guaranteed containment or licence for ignition

WARNING / IMPLEMENTATION / OUTCOME LIMIT

FIRE LINE → maintained break in fuel continuity

PRESCRIBED BURN → authorised planned fuel treatment

HAZARD / ACTOR / INSTRUMENT

- FIRE LINE → maintained break in fuel continuity
- PRESCRIBED BURN → authorised planned fuel treatment

DECISIVE RISK OR MANDATE DISTINCTION

- BOTH → objectives conditions supervision contingency
- NEITHER → guaranteed containment or licence for ignition

WARNING / IMPLEMENTATION / OUTCOME LIMIT

- FIRE LINE → maintained break in fuel continuity
- PRESCRIBED BURN → authorised planned fuel treatment

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NEITHER → guaranteed containment or licence for ignition.

08

STAGE 08 ECOLOGY-SUPPRESSION BALANCE

ECOLOGY-SUPPRESSION BALANCE

WHAT ECOSYSTEM

FIRE REGIME

LIFE OR ASSET THREAT

SAFETY-LED RESPONSE

ECOLOGY OBJECTIVE

BOUNDED MANAGEMENT

UNIVERSAL EXCLUSION OR UNIVERSAL BURNING

WHAT ECOSYSTEM / FIRE REGIME / SEASON / INTENSITY?

LIFE OR ASSET THREAT → SAFETY-LED RESPONSE

FUEL / ECOLOGY OBJECTIVE → BOUNDED MANAGEMENT

TRAP → UNIVERSAL EXCLUSION OR UNIVERSAL BURNING

HAZARD / RISK / RESILIENCE CONCEPT

- WHAT ECOSYSTEM / FIRE REGIME / SEASON / INTENSITY?
- LIFE OR ASSET THREAT → SAFETY-LED RESPONSE

INSTITUTIONAL OR COMMUNITY RESPONSE

- FUEL / ECOLOGY OBJECTIVE → BOUNDED MANAGEMENT
- TRAP → UNIVERSAL EXCLUSION OR UNIVERSAL BURNING

CAPACITY / STATUS / EVIDENCE LIMIT

- WHAT ECOSYSTEM / FIRE REGIME / SEASON / INTENSITY?
- LIFE OR ASSET THREAT → SAFETY-LED RESPONSE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WHAT ECOSYSTEM / FIRE REGIME / SEASON / INTENSITY?.

09

STAGE 09 SMOKE AND INTERFACE MAP

SMOKE AND INTERFACE MAP

FOREST EDGE

HOME ROAD POWER TOURISM SITE

HEALTH MESSAGE + EVACUATION + UTILITY ACTION

RESPONDER AND VULNERABLE-GROUP PROTECTION

MUST REMEMBER

FOREST-FIRE RISK JOINS FUEL, MOISTURE, WEATHER, IGNITION, TERRAIN, FOREST

FIRE → SMOKE / VISIBILITY / TRANSPORT

FOREST EDGE → HOME ROAD POWER TOURISM SITE

HEALTH MESSAGE + EVACUATION + UTILITY ACTION

RESPONDER AND VULNERABLE-GROUP PROTECTION

HAZARD / RISK / RESILIENCE CONCEPT

- FIRE → SMOKE / VISIBILITY / TRANSPORT
- FOREST EDGE → HOME ROAD POWER TOURISM SITE

INSTITUTIONAL OR COMMUNITY RESPONSE

- HEALTH MESSAGE + EVACUATION + UTILITY ACTION
- RESPONDER AND VULNERABLE-GROUP PROTECTION

CAPACITY / STATUS / EVIDENCE LIMIT

- MUST REMEMBER: Forest-fire risk joins fuel, moisture, weather, ignition, terrain, forest...
- FIRE → SMOKE / VISIBILITY / TRANSPORT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Forest-fire risk joins fuel, moisture, weather, ignition, terrain, forest.

09

STAGE 09 SMOKE AND INTERFACE MAP

SMOKE AND INTERFACE MAP

FOREST EDGE

HOME ROAD POWER TOURISM SITE

HEALTH MESSAGE + EVACUATION + UTILITY ACTION

RESPONDER AND VULNERABLE-GROUP PROTECTION

MUST REMEMBER

FOREST-FIRE RISK JOINS FUEL, MOISTURE, WEATHER, IGNITION, TERRAIN, FOREST

FIRE → SMOKE / VISIBILITY / TRANSPORT

FOREST EDGE → HOME ROAD POWER TOURISM SITE

HEALTH MESSAGE + EVACUATION + UTILITY ACTION

RESPONDER AND VULNERABLE-GROUP PROTECTION

HAZARD / RISK / RESILIENCE CONCEPT

- FIRE → SMOKE / VISIBILITY / TRANSPORT
- FOREST EDGE → HOME ROAD POWER TOURISM SITE

INSTITUTIONAL OR COMMUNITY RESPONSE

- HEALTH MESSAGE + EVACUATION + UTILITY ACTION
- RESPONDER AND VULNERABLE-GROUP PROTECTION

CAPACITY / STATUS / EVIDENCE LIMIT

- MUST REMEMBER: Forest-fire risk joins fuel, moisture, weather, ignition, terrain, forest...
- FIRE → SMOKE / VISIBILITY / TRANSPORT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Forest-fire risk joins fuel, moisture, weather, ignition, terrain, forest.

10

STAGE 10 RESTORATION SEQUENCE

RESTORATION SEQUENCE

SAFETY + BURNT AREA

SEVERITY ASSESSMENT

EROSION RUNOFF AND INVASIVE-SPECIES CONTROL

LIVELIHOOD SUPPORT + ECOLOGICALLY SUITABLE RESTORATION

MONITOR RECOVERY

DO NOT EQUATE WITH PLANTING COUNT

CLOSE DISTINCTION

SAFETY + BURNT AREA / SEVERITY ASSESSMENT

EROSION RUNOFF AND INVASIVE-SPECIES CONTROL

LIVELIHOOD SUPPORT + ECOLOGICALLY SUITABLE RESTORATION

MONITOR RECOVERY → DO NOT EQUATE WITH PLANTING COUNT

CLOSE DISTINCTION: Hotspot is not confirmed fire; burnt area is not tree-cover loss

RISK CONDITION / INPUT

- SAFETY + BURNT AREA / SEVERITY ASSESSMENT
- EROSION RUNOFF AND INVASIVE-SPECIES CONTROL

PREVENTION / PREPAREDNESS / RESPONSE MECHANISM

- LIVELIHOOD SUPPORT + ECOLOGICALLY SUITABLE RESTORATION
- MONITOR RECOVERY → DO NOT EQUATE WITH PLANTING COUNT

LAST-MILE / INCLUSION / OUTCOME LIMIT

- CLOSE DISTINCTION: Hotspot is not confirmed fire; burnt area is not tree-cover loss;...
- SAFETY + BURNT AREA / SEVERITY ASSESSMENT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Hotspot is not confirmed fire; burnt area is not tree-cover loss;.

11

STAGE 11 FOREST-FIRE ANSWER SPINE

FOREST-FIRE ANSWER SPINE

CLASSIFY FIRE + TRACE IGNITION WEATHER FUEL TOPOGRAPHY

ADD COMMUNITY FIRE-LINE ECOLOGY SMOKE AND WUI GOVERNANCE

SEPARATE ALERT CAPACITY CONTAINMENT AND RECOVERY OUTCOMES

EVIDENCE LIMIT

NAME SENSOR/PRODUCT, RESOLUTION, DATE

CLASSIFY FIRE + TRACE IGNITION WEATHER FUEL TOPOGRAPHY

PREVENT → DETECT → VERIFY → RESPOND → RESTORE

ADD COMMUNITY FIRE-LINE ECOLOGY SMOKE AND WUI GOVERNANCE

SEPARATE ALERT CAPACITY CONTAINMENT AND RECOVERY OUTCOMES

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name sensor/product, resolution, date

HAZARD / RISK / RESILIENCE CONCEPT

- CLASSIFY FIRE + TRACE IGNITION WEATHER FUEL TOPOGRAPHY
- PREVENT → DETECT → VERIFY → RESPOND → RESTORE

INSTITUTIONAL OR COMMUNITY RESPONSE

- ADD COMMUNITY FIRE-LINE ECOLOGY SMOKE AND WUI GOVERNANCE
- SEPARATE ALERT CAPACITY CONTAINMENT AND RECOVERY OUTCOMES

CAPACITY / STATUS / EVIDENCE LIMIT

- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name sensor/product, resolution, date,...
- CLASSIFY FIRE + TRACE IGNITION WEATHER FUEL TOPOGRAPHY

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name sensor/product, resolution, date,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Name sensor/product, resolution, date,.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

NATIONAL MASTER PLAN FOR FOREST FIRE CONTROL'S FOUR-PART DESIGN

MISSING DEDICATED INSTITUTION

VISIONIAS EXPLICITLY RECOMMENDS

THE FAST 3.0 INSTITUTIONAL CHAIN AS A GENUINE TECHNOLOGY-TRANSFER

THE 44%-DETECTIONS-IN-3%-OF-LAND-AREA CONCENTRATION (PDF P. 47)

VISIONIAS NAMES SEVERAL SPECIFIC RESEARCH/DATA GAPS

VISIONIAS'S OWN CRITIQUE (SECTION 2) IS FUNDAMENTALLY AN EQUITY/

OPTIONAL RISK / RESILIENCE DEPTH

- National Master Plan for Forest Fire Control's four-part design :
- CAUTION: Missing dedicated institution : VisionIAS explicitly recommends
- The FAST 3.0 institutional chain as a genuine technology-transfer
- The 44%-detections-in-3%-of-land-area concentration (PDF p. 47)

MANDATE, WARNING OR OUTCOME LIMIT

- CAUTION: VisionIAS names several specific research/data gaps: "inadequate
- VisionIAS's own critique (Section 2) is fundamentally an equity/
- CAUTION: Fire-suppression cost versus long-term forest-management
- CAUTION: Wildland-urban interface risk (a concept VisionIAS's own

COMPARATIVE HAZARD / GOVERNANCE NUANCE

- NOT: The "FAST 3.0 exists, therefore forest-fire risk is managed"
- CAUTION: This is a long, source-explicit list of named institutional gaps —
- CAUTION: VisionIAS explicitly names "funding" as one of the "missing"
- Burnt Scar Assessment's semi-automated methodology (under FSI

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.